TEXORA · LECTURE SHEET

Technical Fibers and Their Uses

Some fibres are chosen not for comfort but for performance under load, heat, flame or chemical attack. This sheet defines what makes a fibre technical, compares the main high-performance families, and shows why specific — weight-corrected — properties decide their uses.

YM-4203 · TECHNICAL FIBRE & YARN · TOPIC 1 · LEVEL 4 TERM 2 · YARN MANUFACTURING SPECIALIZATION

1. What makes a fibre “technical”?

A **technical fibre**, also called a high-performance fibre, is a fibre engineered so that one or more functional properties are far beyond those of conventional apparel fibres such as cotton, polyester or nylon. The property may be mechanical (very high tenacity or modulus), thermal (survival at high temperature, non-flammability), chemical (resistance to acids, solvents or ultraviolet light), or electrical. A technical textile is then a textile designed primarily for that function rather than for decoration.

No single property makes a fibre technical. A useful working test is to ask four questions:

- **How far beyond the ordinary is the key property?** A para-aramid, for example, is roughly twice as strong as nylon at about half the breaking extension, and it does not melt.
- **Does the property survive the service environment?** Strength at room temperature is useless in a fire suit if the fibre softens at 180 °C, or in a marine rope if it creeps under sustained load.
- **Can the fibre be processed?** It must be spinnable or drawable into a continuous filament, then twistable, weavable, braidable or needlable into a usable form.
- **Is the performance worth the cost?** Technical fibres are expensive, often many times the price of commodity fibres, so they are used where failure is dangerous, where weight saving pays, or where no ordinary material survives.

> **KEY IDEA**
>
> Technical fibres are selected as a **package**: strength, stiffness, density, temperature limit, chemical resistance, durability and price together. Choosing on tenacity alone is the most common beginner's error.

2. The language of comparison

2.1 Tenacity, modulus and breaking extension

Tenacity is breaking load divided by linear density, commonly reported in cN/dtex or N/tex for fibres, and as tensile strength in GPa for engineering comparison. **Initial modulus** is the stiffness at small strain — the slope of the first, straight part of the stress–strain curve. A high-modulus fibre stretches very little under load, which matters in composites, ropes under precise length control and reinforcement. **Breaking extension** is the strain at failure: low for carbon and glass (brittle behaviour), higher for UHMWPE and nylon (more forgiving, more energy-absorbing).

2.2 Specific properties — why density matters

In aircraft, ropes, cables and body armour, weight itself is a load. Engineers therefore divide strength or modulus by density to obtain **specific tenacity** and **specific modulus**:

$$\text{Specific tenacity} = \text{tensile strength (GPa)} / \text{density (g/cm}^3\text{)}, \text{ in GPa·cm}^3\text{/g}$$

A fibre of moderate strength but very low density can outperform a stronger, heavier material on this measure. This is why UHMWPE, at a density below that of water, dominates high-performance cordage despite a modest melting point, and why steel wire — strong in absolute terms — scores poorly on a weight basis.

2.3 Thermal behaviour: LOI, melting and decomposition

Three different ideas must not be confused. The **limiting oxygen index (LOI)** is the minimum percentage of oxygen in an oxygen–nitrogen mixture that will just support combustion; air contains about 21% oxygen, so a fibre with LOI well above 21 tends to self-extinguish in air. The **melting point** applies only to thermoplastic fibres. Many high-performance fibres — aramids, carbon, PBI — do not melt at all; they **decompose** or char at high temperature, which is often more useful, because a charred layer insulates and the fabric does not drip. The **continuous-use temperature** is lower than the decomposition temperature: it is the temperature at which the fibre can work for long periods without serious loss of strength.

3. The main families

All values in this section and in Table 1 are **standard textbook values — typical ranges for commercial grades**. A specific commercial grade must always be checked against its manufacturer's data sheet before design use.

3.1 Para-aramid (e.g. Kevlar, Twaron)

An aromatic polyamide with para-linked amide groups, giving a straight, highly oriented, liquid-crystal-spun molecule. Very high tenacity and modulus, no melting, decomposition around 500 °C, LOI about 29. Weaknesses: poor UV resistance, low compressive strength, and moisture uptake of a few percent. Uses: ballistic vests, cut-resistant gloves, tyre cord, conveyor belts, composites.

3.2 Meta-aramid (e.g. Nomex)

The meta-linked sister fibre: less oriented, so tenacity is much lower, but outstanding flame and heat resistance, LOI about 28–30, no melting, and good electrical insulation. It is the fibre of fire-service turnout gear, racing suits, hot-gas filter bags and electrical insulation paper.

3.3 Carbon fibre

Made by stabilising and carbonising a precursor (usually PAN) at very high temperature. Extremely high modulus and high strength, density about 1.8 g/cm³, no melting, surviving very high temperatures in inert atmospheres — but it oxidises in air above roughly 400–500 °C, is conductive, and fails brittlely at only 1–2% extension. Uses: aerospace composites, pressure vessels, wind-turbine blades, sports goods.

3.4 Glass fibre (E-glass)

Drawn from molten glass. Cheap for its strength, non-flammable, chemically durable and an electrical insulator, but dense (about 2.5 g/cm³), brittle, and prone to strength loss from surface damage and moisture at the fibre–resin interface. Uses: glass-reinforced plastics, insulation, filter media, printed-circuit boards.

3.5 UHMWPE (e.g. Dyneema, Spectra)

Ultra-high-molecular-weight polyethylene, gel-spun and ultra-drawn (Topic 3). The highest specific tenacity of any commercial fibre, density about 0.97 g/cm³ so it floats, excellent abrasion and chemical resistance — but it melts near 145–150 °C and **creeps** under sustained load. Uses: marine and mooring ropes, ballistic protection, cut-resistant gloves, medical sutures.

3.6 PTFE, PBI and basalt

PTFE is chemically almost inert, continuously usable to about 260 °C, with very low friction, but low strength and severe creep — used in chemical filter media, gaskets and membranes. **PBI** (polybenzimidazole) is among the most flame-resistant organic fibres, LOI around 41, no melting, used in demanding fire-blocking applications where its high price is justified. **Basalt**, drawn from melted volcanic rock, is non-flammable with good heat resistance — used in fire barriers, insulation and some composites.

FIBRE	TENACITY (GPA)	MODULUS (GPA)	DENSITY (G/CM³)	THERMAL BEHAVIOUR	KEY USES
Para-aramid	2.8–3.6	60–120	1.44	No melt; decomposes ~500 °C; LOI ~29	Ballistics, tyre cord, gloves
Meta-aramid	0.4–0.7	7–12	1.38	No melt; LOI 28–30	Fire suits, hot-gas filters
Carbon (PAN)	3.0–5.0	230–400	1.75–1.85	No melt; oxidises in air >~450 °C	Aerospace composites
E-glass	2.0–3.5	70–75	2.54	Non-flammable; softens ~840 °C	GRP, insulation, PCB
UHMWPE	3.0–4.0	100–170	0.97	Melts ~145–150 °C; LOI ~18	Ropes, ballistics, gloves
PTFE	0.2–0.4	~1–2	2.1–2.2	Continuous use to ~260 °C; LOI ~95	Chemical filters, membranes
PBI	0.3–0.4	~5–6	1.43	No melt; LOI ~41	Fire blocking, fire service
Basalt	3.0–4.8	85–95	2.6–2.8	Non-flammable; use to ~600 °C	Fire barriers, insulation

Table 1 — Typical textbook ranges for commercial grades; verify a specific grade before design use. Note how no fibre wins every column: selection is always a compromise.

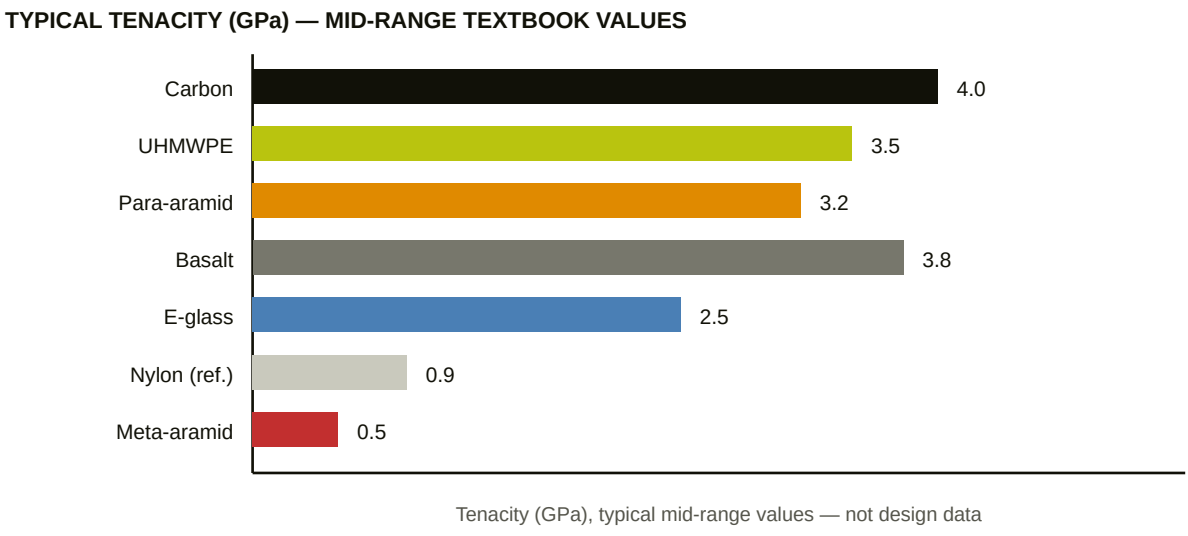


Figure 1 — Typical tenacity comparison. High tenacity alone does not make a fibre the right choice: meta-aramid is weak on this chart yet irreplaceable in fire protection.

4. Worked example — specific tenacity decides

WORKED EXAMPLE — CARBON FIBRE VERSUS STEEL WIRE

Carbon fibre: tensile strength 3.5 GPa, density 1.8 g/cm³.

Specific tenacity = 3.5 / 1.8 = **1.94 GPa·cm³/g**.

Steel wire: tensile strength 1.8 GPa, density 7.8 g/cm³.

Specific tenacity = 1.8 / 7.8 = **0.23 GPa·cm³/g**.

Ratio = 1.94 / 0.23 ≈ **8.4**. Weight for weight, the carbon fibre carries about eight times the load of the steel wire — which is why a carbon or UHMWPE rope can replace a far heavier steel wire rope in weight-critical lifting and mooring, provided temperature, abrasion and creep limits are respected.

The comparison is only honest when service conditions are stated with it: steel keeps its strength at temperatures that would melt UHMWPE, and it does not creep. Specific properties rule *inside an operating envelope*, not in isolation.

5. Selection framework

1. **State the function** — carry load, block flame, filter hot gas, insulate, conduct, resist cutting.
2. **State the environment** — temperature (peak and continuous), chemicals, UV, moisture, abrasion.
3. **Eliminate on hard limits first** — a fibre that melts, decomposes or creeps in the environment is out, however strong it is.
4. **Compare the survivors on specific properties and durability**, then on processability into the required form (woven, braided, nonwoven, composite).
5. **Check cost per unit of performance over the service life**, including replacement — a cheap fibre replaced often may cost more than an expensive one that lasts.

Glossary

Technical fibre — fibre engineered for functional performance far beyond conventional fibres. **Tenacity** — breaking load per unit linear density. **Modulus** — stiffness, the slope of the initial stress–strain curve. **Specific tenacity** — tensile strength divided by density. **LOI** — limiting oxygen index, the minimum oxygen percentage supporting combustion. **Decomposition** — chemical breakdown at high temperature in fibres that do not melt. **Creep** — continuing extension under sustained load. **Aramid** — aromatic polyamide fibre. **UHMWPE** — ultra-high-molecular-weight polyethylene.

Key points

• A technical fibre is judged as a package of mechanical, thermal, chemical and durability properties, never on tenacity alone.
• Specific tenacity = strength / density; low-density fibres win where weight is itself a load.
• LOI, melting and decomposition are different ideas — many top fibres do not melt but char or decompose.
• Para-aramid = strength + heat; meta-aramid = flame protection at modest strength; carbon = stiffness; UHMWPE = specific strength, but melts ~150 °C and creeps.
• All property values here are typical textbook ranges — confirm the grade before design use.
• Selection order: function → environment → hard limits → specific properties → lifetime cost.

Practice

1. Calculate the specific tenacity of a para-aramid of tensile strength 3.0 GPa and density 1.44 g/cm³, and of E-glass of 2.0 GPa and density 2.54 g/cm³. Which is better weight-for-weight, and by what ratio?
2. A hot-gas filter must work continuously at 220 °C in air. Using Table 1, eliminate the unsuitable fibres and justify a shortlist of two.
3. Explain why LOI alone cannot be used to declare a fibre “fireproof”, referring to decomposition and fabric construction.
4. A mooring rope is to be as light as possible but will sit loaded for months. Discuss the UHMWPE creep problem and name one alternative fibre, with its trade-off.

Answer key

1. Para-aramid: 3.0 / 1.44 = **2.08 GPa·cm³/g**. E-glass: 2.0 / 2.54 = **0.79 GPa·cm³/g**. Para-aramid is better weight-for-weight; ratio 2.08 / 0.79 ≈ **2.6**.
2. UHMWPE melts near 150 °C — out. Carbon oxidises and is brittle for a filter felt — out. Shortlist: **meta-aramid** (no melting, flame resistant, the classic hot-gas filter fibre) and **PTFE** (continuous use to ~260 °C, chemically inert), choosing on chemical exposure and cost.
3. LOI measures only whether a small specimen keeps burning in a controlled oxygen mixture. It says nothing about heat release, char strength or shrinkage in a real fire; a high-LOI fibre can still decompose and lose strength at temperature. “Flame resistant” under stated test conditions is the honest claim, not “fireproof”.
4. UHMWPE creeps: under sustained load the rope permanently elongates, changing tension and length. Alternative: **para-aramid** — much lower creep and better heat resistance, at the cost of UV sensitivity (needs a jacket), higher density (1.44 vs 0.97, so it sinks) and higher price.